ONE PERSON. SIX MODALITIES. FIVE LAYERS.

Henry Zhang — one person, *full stack*.

Independent researcher. Hangzhou. Currently at ModelBest / MiniCPM. The matrix on page 3 is what the work looks like when the loop on page 2 runs continuously for a decade.

> *Theory → Tools → Runtime → Product → Culture.*
> *The whole stack, owned.*
>
> — SELF-DESCRIPTION. THE EVIDENCE HOLDS UP.

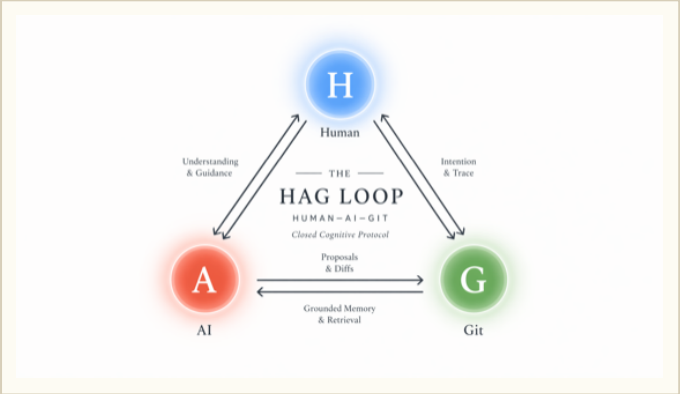
PREPRINTS	PUBLIC REPOS	SHIPPED	AUDIENCE
2 Zenodo DOIs	**19 · GitHub**	**APK · macOS · iOS · visionOS**	**Bilibili 100 k view**
TRB · SFP V2	LLM · VLM · VOICE · EDGE	5 SILICON VENDORS	SPACE.BILIBILI.COM/188066555

THE ENGINE ROOM

§01 How the work gets built.

The matrix two pages from now isn't a persona. It isn't a résumé either. It's what a single feedback loop outputs when it runs continuously, across modalities, across layers of the stack, for ten years.

The loop has three moving parts. Strip away any one and the matrix does not materialize.



H — **Human.** Asks, and doesn't stop at the first plausible answer. Most of the signal lives in the second and third question.

A — **AI.** Pattern-completes. Surfaces adjacencies the human would not have reached alone. Fast, wide, occasionally wrong.

G — **Git.** Append-only memory. Branches when curiosity forks; rolls back when a branch turns out wrong. Every artifact on the next page has a commit hash attached.

§ Why it compounds

Each loop iteration produces two things: a *shipped artifact* (code, paper, APK, an audio demo), and a *structural lesson* about where the next loop should start. The artifacts are what the outside world sees. The lessons are what make the next artifact cheaper than the previous one.

That's why the output spans six modalities without looking diluted: by the time the loop reaches *ATC speech*, the lessons from *representation bandwidth*, *satellite VLM*, and *edge NPU deployment* have already collapsed most of the dead ends.

> WHAT FOLLOWS
>
> The matrix is a six-by-five grid. Rows are modalities (*LLM, Vision, Speech, Edge, Agent, Meta-tooling*). Columns are layers (*Science, Tools, Runtime, Product, Content/IP*). Every cell links to a real repo, a real DOI, or a real shipped binary. If a cell is empty, it's because nothing shipped yet — not because the slot is aspirational.

THE OUTPUT

§02 The Matrix — what actually ships.

Six modalities × five stack-layers. Every cell is a link to real code, a real DOI, or a real shipped binary. An em-dash means the slot is genuinely empty — not "coming soon".

	SCIENCE	OPEN-SOURCE TOOLS	RUNTIME / MODELS	PRODUCTS / DEPLOY	CONTENT / IP
LLM · Text	**TRB** (P1) · Representation Bandwidth **SFP v2** (P2) · probe kit rl-drift	**spectral-flow-probe** · 20-min RL scan turboquant-pytorch · ICLR'26 RepEngvLLM	nano-vllm + TurboQuant · 5× KV flash-attn-blackwell · sm_120	claude-code fork open-code-agent	superpowers-cn · 10 k★ · CN
Vision · VLA	**RepSNI** · 14 models, SNI TRB §VLA · geometry	SDE · DarkSpace Joi · drift engine	minicpmv_onnx multi-image async VLM (QNN)	**AVIA — Satellite VLM** libtorch C++ · on-board	**Seedance 2.0 Planner** 44★ · ByteDance video
Speech · Audio	—	dereverb_audio · DSP	VoxCPM · voice-clone Omni · MiniCPM-o 4.5 gguf	**AVIA — ATC ASR** · Apple Silicon RTF > 3 · mem < 2 GB CallCenter-VA · Hybrid-VA · Voiceagent-Mac	**PixelSynesthesia** · music→pixel «枪兵颂» AI MV · 100 k · 7.1 k
Edge · Hardware	MiniCPM cross-platform bench	—	QNN · MTK · RKNN Intel NPU · Apple Silicon	EdgeClaw-audit multi-tenant isolation	—
Agent · Workflow	**HAG Loop** · Human–AI–Git	ScalEdgeClaw	cursor-rts-audio	**PocketClaw** APK · 20★ · 19.4 k Kotlin · no server OmniAgent · Swift	**OpenMAIC** · THU MAIC · voice
Meta · Tooling	—	**pgattn** · LM-native PDF ↔ git *"vLLM pages KV; pgattn pages attention."*	wechat-extract · chat ETL	html-inbox · doc staging	五代风云 · Five Dynasties

IF YOU'RE HERE FOR ONE THING

§03 Read these two papers first.

Both Zenodo preprints. Both reproducible on a single consumer GPU. One thesis: **RL alignment is a geometric rotation, not a capacity loss** — and a tool that measures it.

Paper 1 · *The Representation Bandwidth* (TRB)

CONSERVATION ANALYSIS · RL ALIGNMENT · N = 644

RL alignment is widely believed to compress the representational capacity of a pretrained transformer. **It does not.**

Across Qwen / Yi / Mistral base ↔ instruct pairs, the singular-value spectrum of every weight matrix drifts below **0.5%**, while the singular-vector bases rotate by **0.9–8°**. Alignment is consistent with a **graph isometry**: spectrum preserved, bases rotated. What people call "PR collapse" is not capacity loss — it is a measurement artifact of angle.

Pearson $p = 7 \times 10^{-13}$.

DOI · 10.5281/zenodo.19626829

Paper 2 · *Spectral Flow Probe v2* (SFP)

MEASUREMENT TOOLKIT · 20 GPU-MIN / 7 B PAIR · MIT

Five components implementing the TRB measurement protocol as a reusable library:

- **SpectralProbe** — deterministic 7-band bandwidth.
- **RotationAnalyzer** — weight-space SVD & principal angles.
- **BandwidthDiagnostic** — pre-training data-mix audit.
- **SpectralCallback** — training-time monitor.
- **spectral_pr_loss** — differentiable per-band regularizer.

Every measurement runs on a single consumer GPU. Reproducible to 10^{-10} across hardware.

DOI · 10.5281/zenodo.19587024 · github.com/HenryZ838978/spectral-flow-probe

WHERE THE WORK CONNECTS

§04 Partnerships, by open-source home.

Not sponsorships — public-code intersections. Every name below is an org whose open-source repos overlap directly with page 3.

QUALCOMM · MEDIATEK · ROCKCHIP · INTEL · APPLE · HUAWEI · BYTEDANCE · OPENBMB · MINICPM · THU MAIC

Silicon side covers every major non-NVIDIA on-device inference target; OpenBMB is the primary research collaboration; Huawei & ByteDance ecosystems are where voice and video work deploys; THU MAIC hosts *OpenMAIC-VoiceSupport*.

SIGNAL OUT

§05 Track record.

Zenodo preprints	2 DOIs (TRB + SFP v2) · download-to-view ratio 63 % and 87 %.
Public GitHub repos	**19** — spanning LLM, VLM, voice, mobile, edge hardware.
Shipped products	Android APK (`PocketClaw v0.4.1`), macOS / iOS / visionOS voice agents, MiniCPM on **5 silicon vendors** (Qualcomm, MediaTek, Rockchip, Intel, Apple Silicon).
Content reach	Bilibili · space.bilibili.com/188066555 · single AI MV at **100 k view / 7.1 k engagement**.
Culture IP	五代风云 (*Five Dynasties*) — long-running historical-AI fusion project. Not a side quest; it's the Content/IP column of the matrix.

§ About this page

If "one-person full-stack AI lab" sounds implausible — and you suspect most of it is filler — the fastest way to find out is to open the repos on page 3. Each cell links to code, a DOI, or a shipped binary. Em-dashes are where nothing ships; they are not placeholders.

If any of it overlaps with what you're building, the mailbox below is monitored. First-email bar is: one specific cell on page 3, one question sharper than *"tell me more"*.

> WHY THIS FILE EXISTS
>
> This document is itself dog-food for the pgattn stack — authored in a single self-contained HTML, styled by `pgattn-report-style`, paginated by `pgattn-pageRestrict`, and exported through the `pgattn` CLI. The report you are reading is a *proof of stack coherence* as much as it is a personnel file.

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